

Stuff vs. People

Ideological Perceptions and Dimensions of Candidate Rhetoric

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The stuff vs. people team



Amanda Loehrke



Max Welch



Chris Hare

“My dream construction set”



Tweet



Dr. Mehmet Oz
@DrOz

I had an informative discussion w/ [@schlouchinc](#) in Berks Co. about the worst supply chain issues the company has experienced in 39 years. We need to stop Biden’s war on American energy and allow our communities & businesses to thrive again. I’ll be a voice for you in the Senate!



4:18 PM · 6/16/22 · [Twitter for iPhone](#)

10:29 AM · Jun 17, 2022



Tweet



John Fetterman
@JohnFetterman



This Tweet was deleted by the Tweet
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What do candidates say, and what do voters hear?

Candidate rhetoric

CampaignView dataset Case 2025;
Porter, Case, and Treul 2025)

Embeddings from candidates'
campaign websites from
2018–2022

Voter perceptions

Voter perceptions (Hassell,
Heseltine, and Reuning 2025)

Liberal Moderate Conservative

Campaign text and perceived ideology

CampaignView

Issue-page text

300-dimensional Doc2Vec embeddings

WEB = first principal component

4,368

candidate-years with embeddings

Voter perceptions

CES ideological placements

Bayesian Aldrich–McKelvey scaling

Higher scores = more conservative

1,860

candidate-years in baseline models

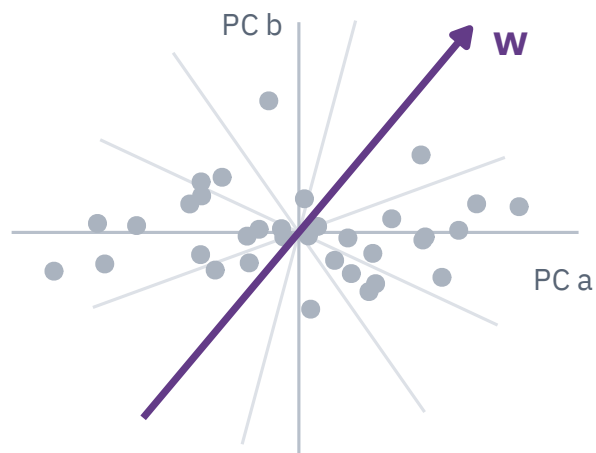
U.S. House campaigns, 2018, 2020, and 2022. Candidate-year unit of analysis.

Three ways to choose the projection

Same embedding space; different roles for voter perceptions

PCA rotation

Rotate within a PC subspace



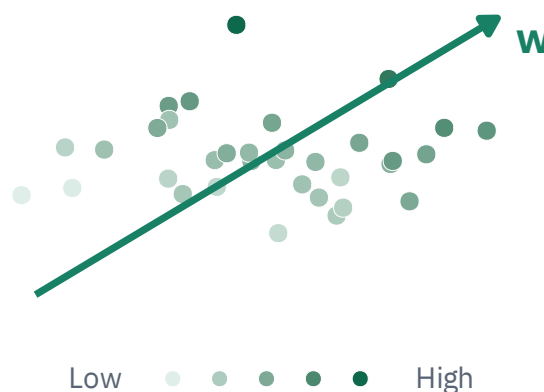
Screen PCs by gender;
compare candidate rotations

Perceptions select

Alongside gender separation

PLS

Fit to residual perceptions



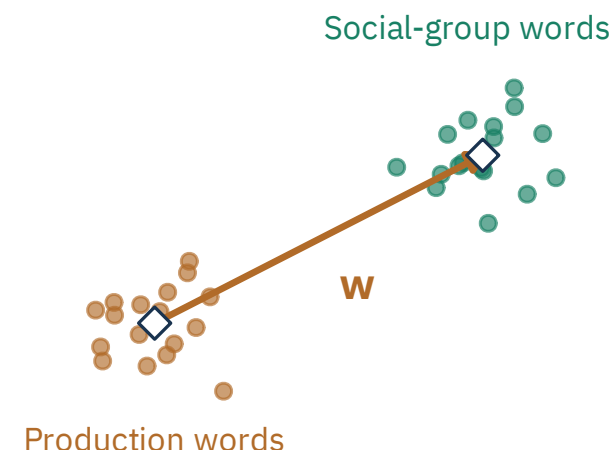
Perceptions remaining after
WEB and controls

Perceptions estimate

Five-fold cross-fitting

Lexicon

Contrast word-set centroids

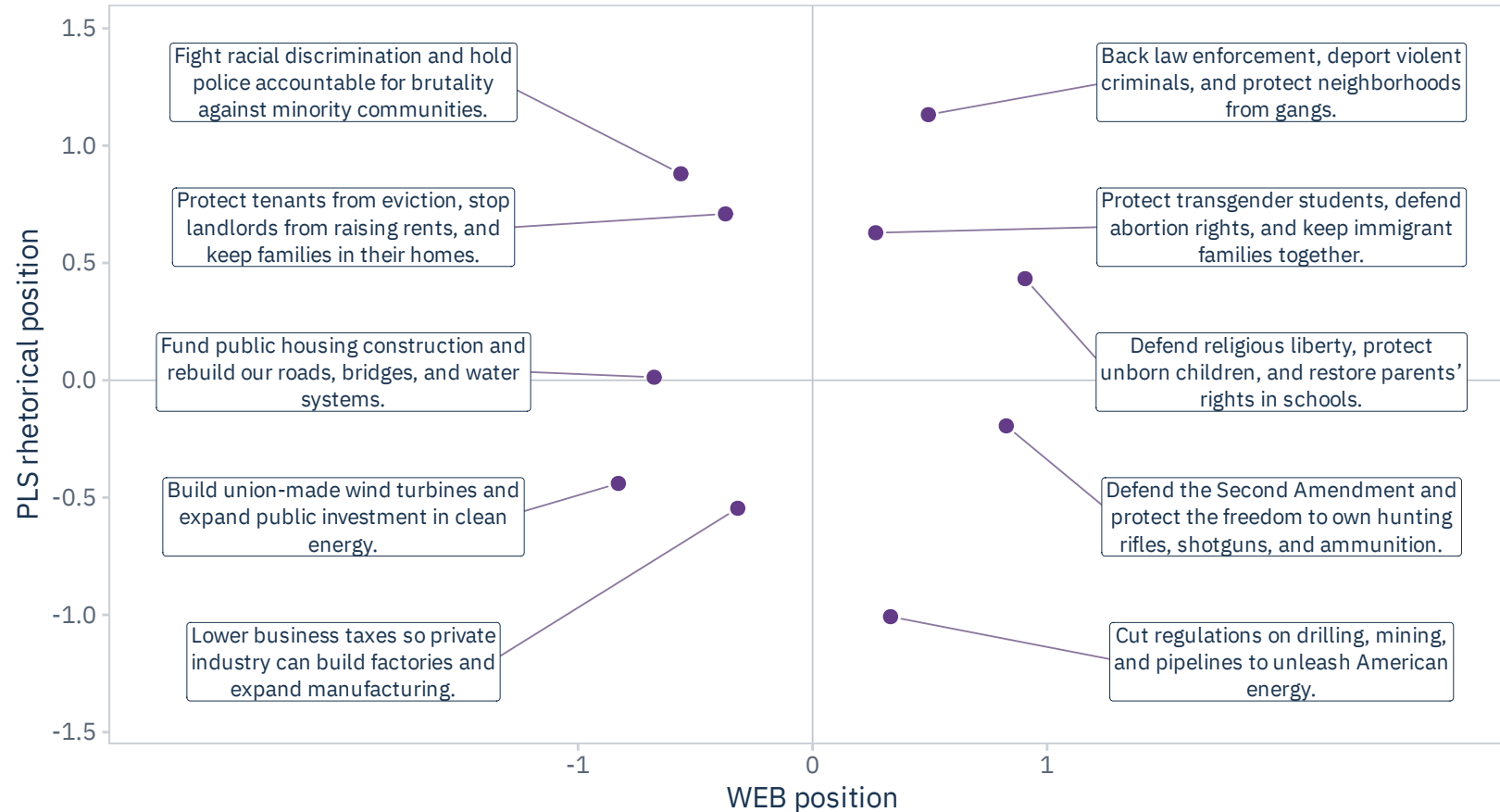


Expand two word sets;
join their mean locations

No perceptions used

In constructing the direction

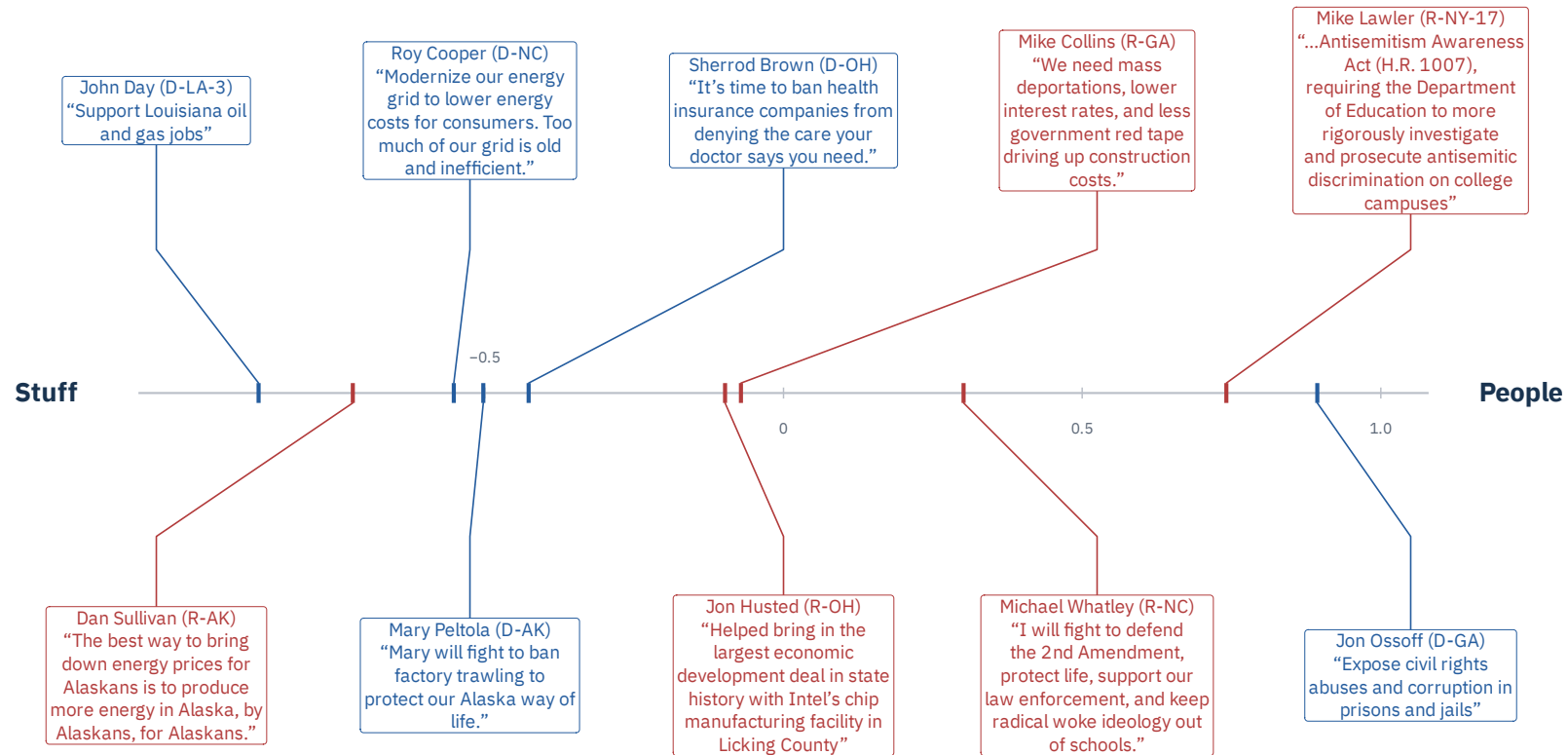
One validation strategy: out-of-sample synthetic rhetoric



Synthetic statements' average word projections; pooled WEB-PLS word-map units.
Equal weight per retained word. Negative WEB = left; positive WEB = right.

Fixed synthetic examples; pooled word-map averages.

Another validation strategy: 2026 candidates



PLS rhetorical position: average word projections of excerpts from 2026 House and Senate campaign websites.
 Democrats are blue; Republicans are red. Two House excerpts were selected after scoring to illustrate the ends of the fixed axis.

2026 House and Senate candidate webpages; pooled word-map averages.

Words defining the lexicon

Production pole (100 words)

Technical work (seeds; 22 of 31)	<i>build, make, manufacture, manufacturing, produce, drill, harvest, machine, construct, construction, skill, skilled, trade, trades, factory, industry, industrial, engineer, engineering, labor, craft, worker</i>
Sectors and materials (seeds; 32 of 34)	<i>oil, gas, coal, mining, minerals, timber, steel, farm, farms, crops, livestock, cattle, corn, pipeline, pipelines, drilling, roads, bridges, highways, ports, rail, grid, broadband, equipment, plant, plants, factories, goods, materials, shipping, freight, trucks</i>
Added by expansion (46)	<i>production, fertilizer, energy, refineries, batteries, infrastructure, renewable, logistics, electric, solar, fuel, products, vehicles, turbines, wind, transportation, industries, electrical, aluminum, hydrogen, diesel, turbine, truck, appliances, projects, technology, regenerative, carbon, electricity, supply, desalination, electronics, railways, liquid, rails, supplies, producing, store, biofuels, fiber, efficiency, greenhouse, fuels, technologies, buildings, biomass</i>

People pole (100 words)

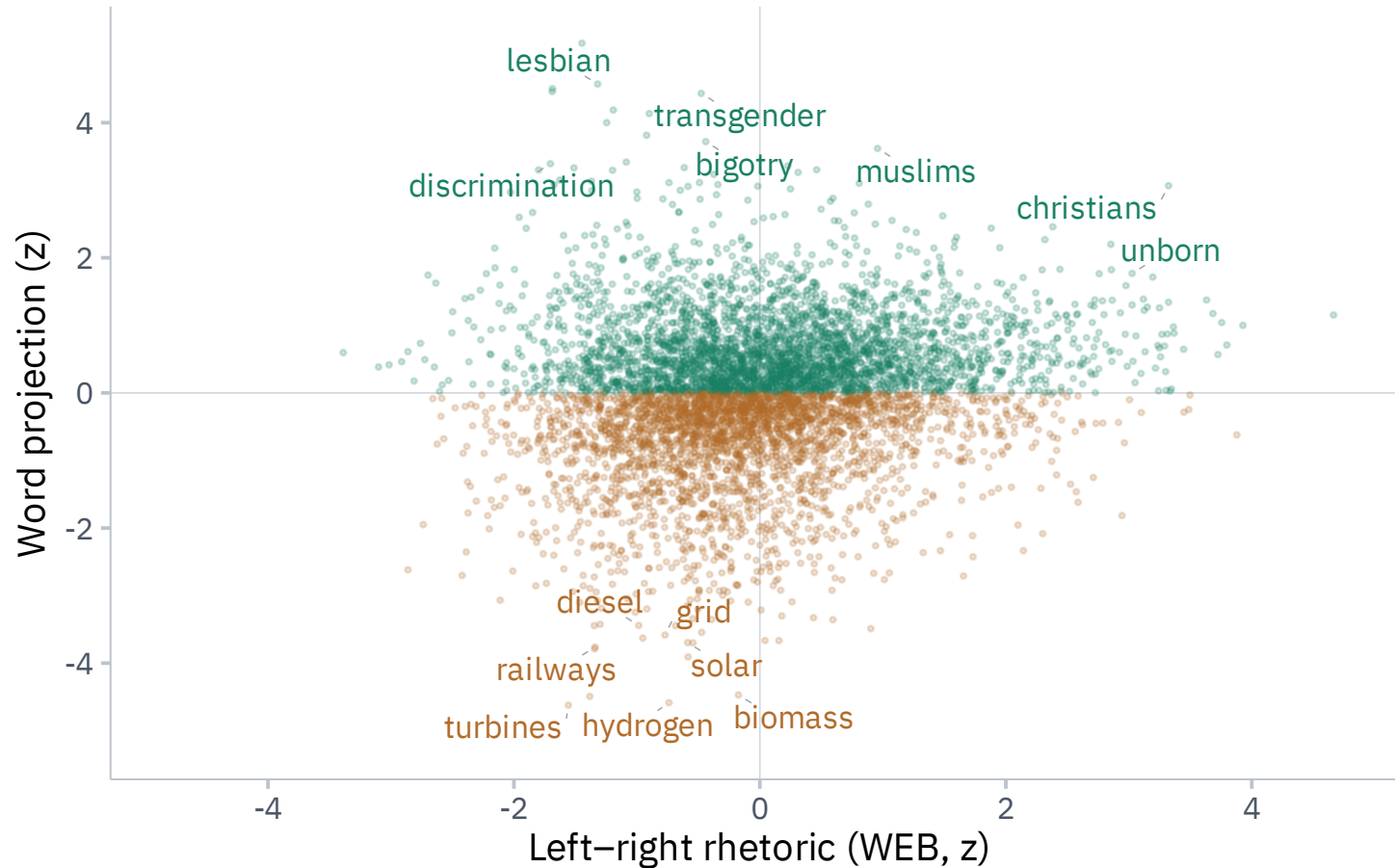
Social groups (seeds; 26 of 26)	<i>immigrants, immigrant, refugees, refugee, migrants, undocumented, dreamers, women, lgbtq, gay, lesbian, transgender, black, african, latino, latinos, latina, hispanic, muslim, muslims, jewish, jews, asian, indigenous, minorities, minority</i>
Added by expansion (74)	<i>lgbt, lgbtqia, latinx, bisexual, queer, conforming, color, trans, discrimination, xenophobic, white, equality, faiths, bigotry, brown, persecuted, persecution, disabled, harassment, xenophobia, disenfranchised, homophobia, marginalized, murdered, native, binary, oppressed, intersex, violence, detained, orientation, hate, mistreatment, religious, imprisoned, female, acknowledges, jailed, unarmed, gender, christians, sexual, ethnic, condemns, hispanics, incarcerated, brutality, hateful, disproportionate, affirm, despicable, disenfranchise, disgraceful, juveniles, voiceless, unjust, discriminatory, louder, disproportionately, amicus, innocent, bullying, predominantly, victims, unhoused, deportations, statistic, genders, disenfranchisement, prejudice, disrespect, hiv, discriminates, institutionalized</i>

Seeds that reach neither pole (11)

Below the frequency floor	<i>builder (9), weld (6), welder (12), machinist (7), mechanic (14), labour (0), craftsman (0), operator (13), technician (17), refinery (12), machinery (24)</i>
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Table: The words that define the lexicon-only direction, by pole and source. The direction is the centroid of the 100 production words minus the centroid of the 100 social-group words. The technical-work seeds translate the technical work logic of the occupational literature Oesch:2006a,Kitschelt:2014a into campaign vocabulary; the sector and material seeds were added to extend the pole from skilled trades to the material economy; the social-group seeds take the place of the care vocabulary that the work-logic literature would suggest. Words added by expansion are the corpus words that best discriminate between the two seed centroids, listed in rank order. Seeds used in fewer than 25 of the 4,368 platforms (document frequency in parentheses) are dropped before the expansion.

The lexicon recovers a clear contrast



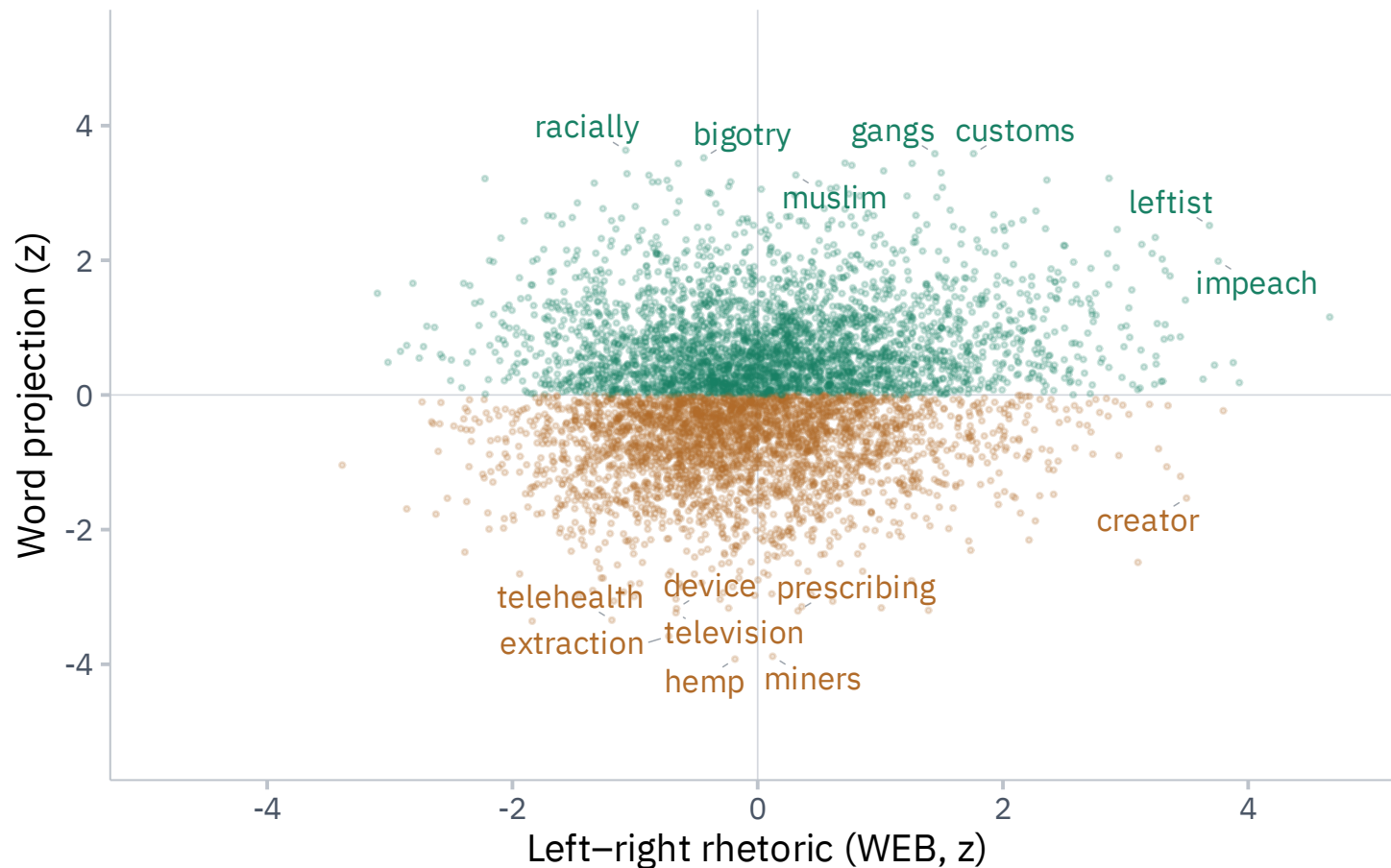
**Social groups
and discrimination**

**Energy and
transportation**

Open interactive map

6,469 words. Projections standardized over the vocabulary. Selected labels.

PLS recovers a less tidy contrast



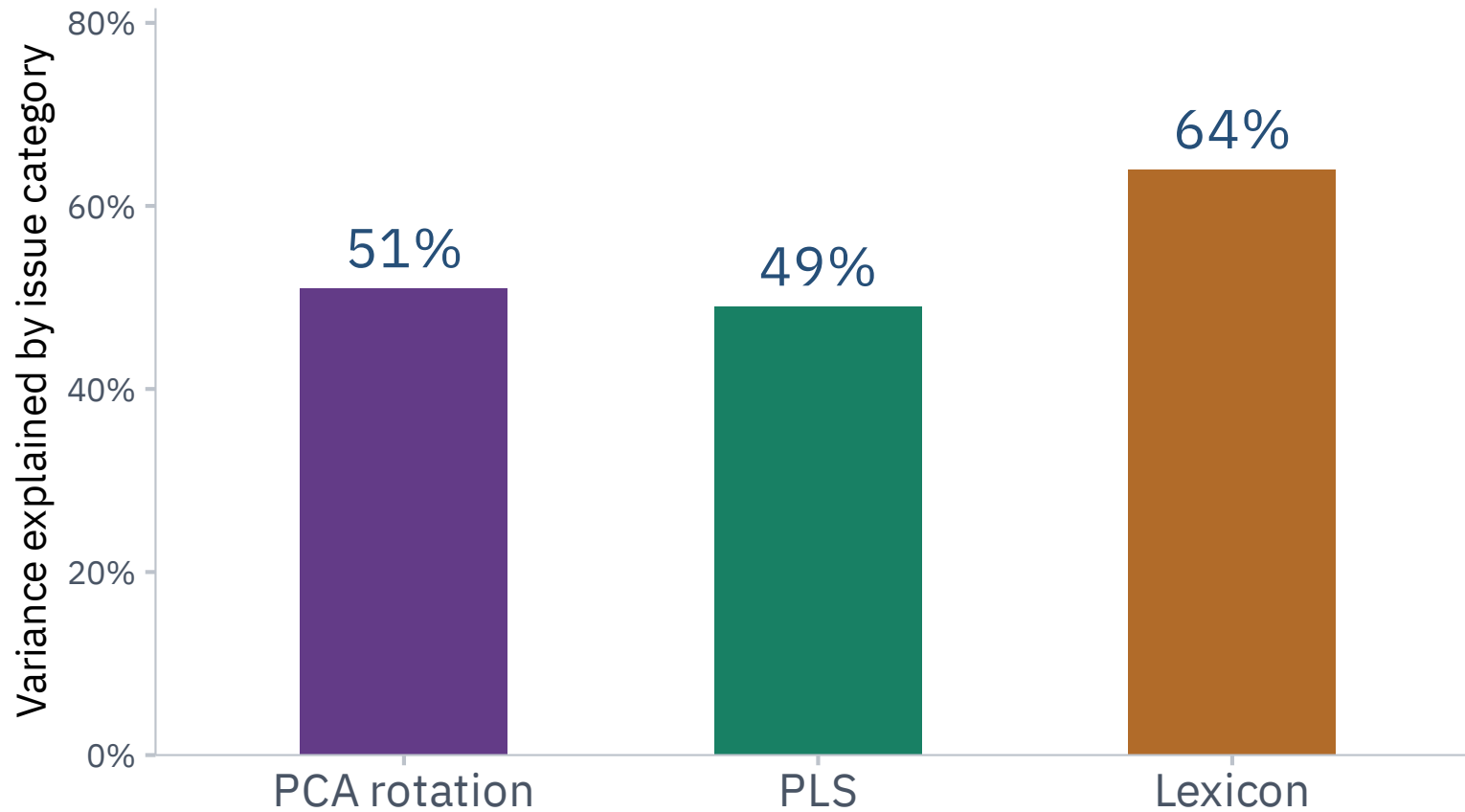
**Groups, conflict,
and policing**

**Production,
health
administration,
and other vocabulary**

Open interactive map

Pooled PLS direction. Same vocabulary and horizontal axis as the preceding slide.

Issue emphasis explains much of each score

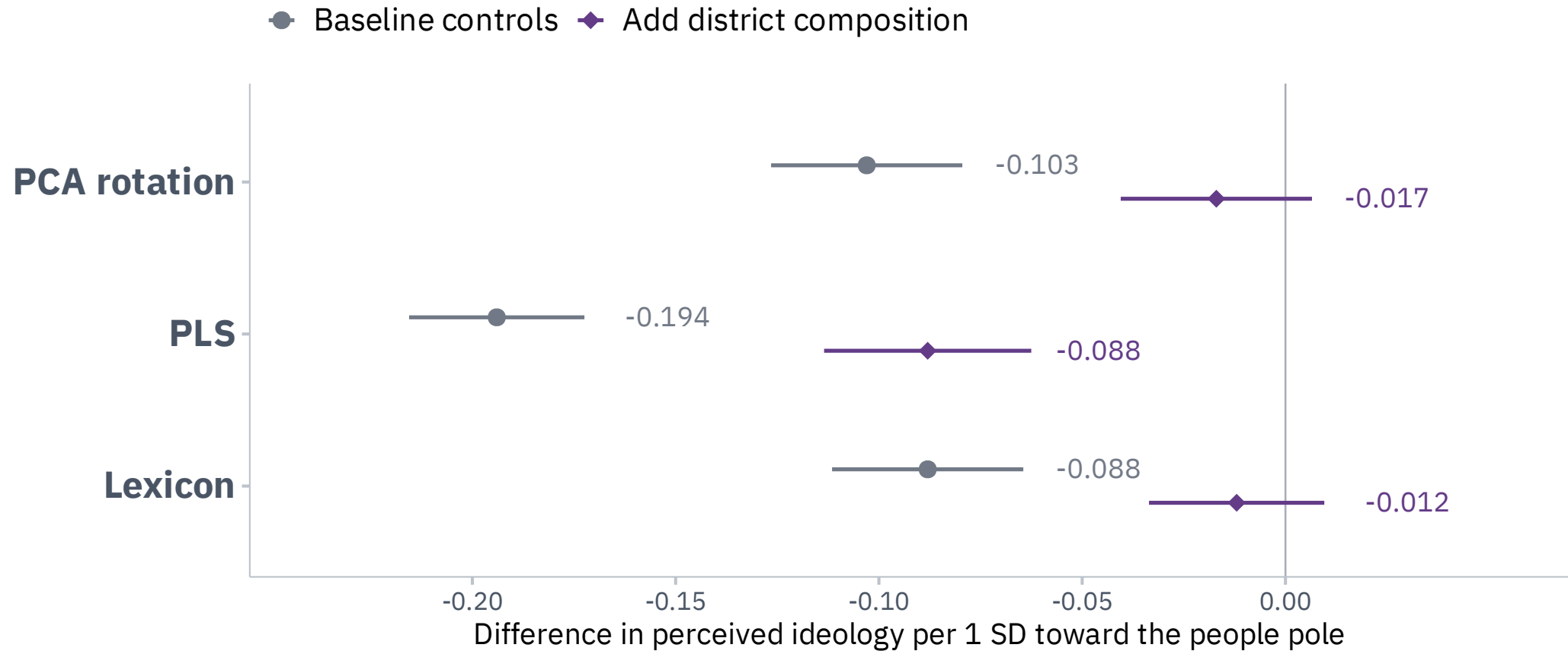


**A large contribution
from issue selection**

Residual variation
requires interpretation

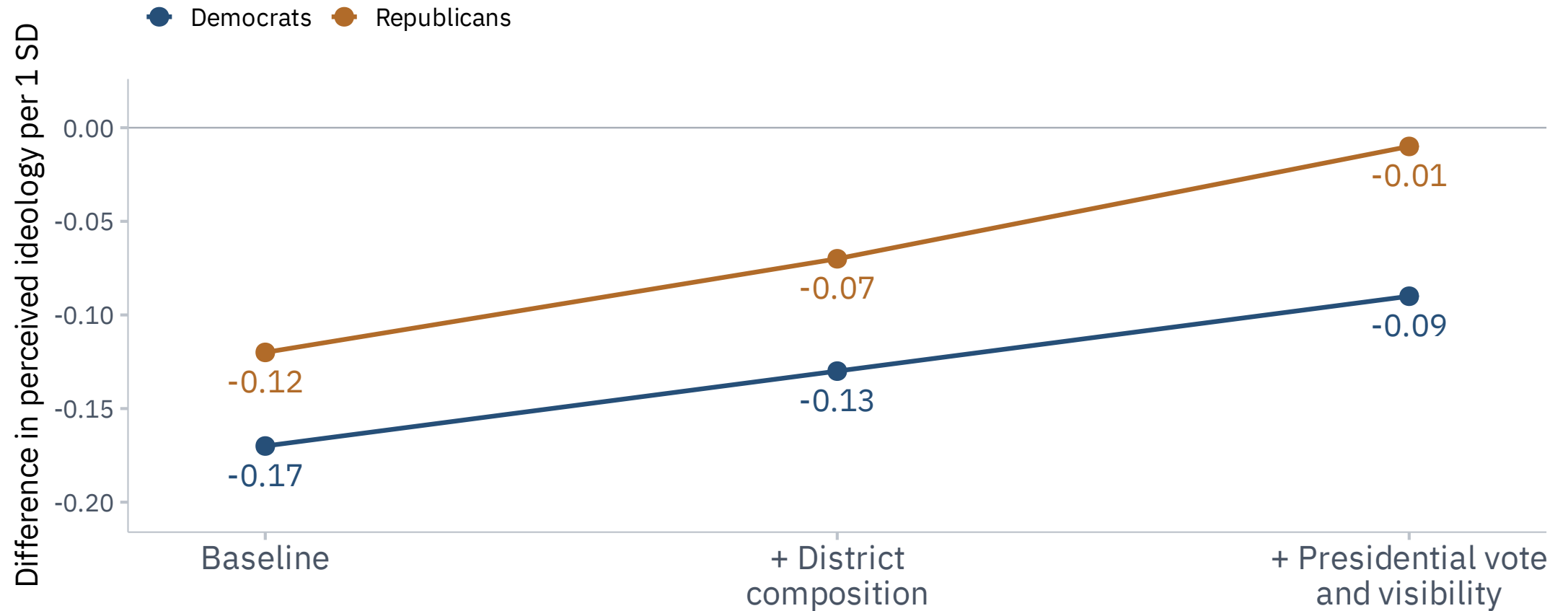
40,073 issue statements. R^2 from issue-category differences in statement scores.

District controls separate the three strategies



95% HC1 intervals. $N = 1,860$ baseline; 1,855 with district composition. Negative = more liberal.

The association differs by party



Party-specific PLS. Rounded point estimates from the paper. Samples vary across specifications.

What the evidence supports

Information beyond conventional left–right rhetoric

The clearest semantic contrast loses its association after district controls

The more persistent PLS association comes with a less coherent substantive label

Conditional associations. Website text measures campaign presentation, not observed exposure.

Next steps

1. Blind coding of words and campaign passages
2. Fresh election cycle and alternative embeddings
3. Experiment with rhetoric, holding policy fixed

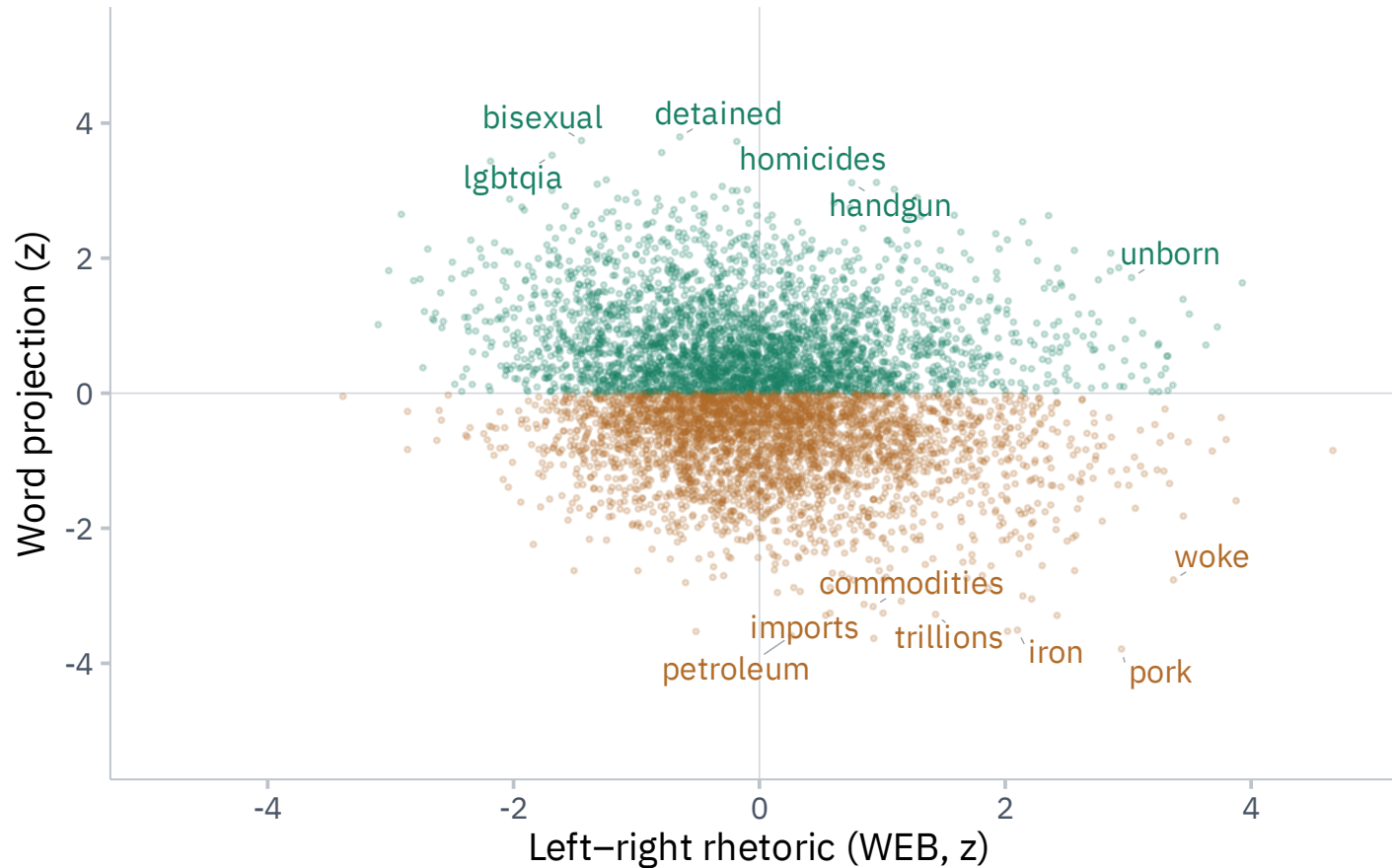
Explore the interactive word map

christopherdanielhare.github.io/candidate-embeddings-word-map.html

[Backup slides](#)

[Interactive visualization: Christopher Hare's GitHub page](#)

PCA search and its selection rule



20 PCs ranked by gender

10 retained

1,620 rotations searched

Selected:
PC1 + PC9 at 105°

$|r \text{ with WEB}| < 0.5$

PCA basis from text. Gender and perceptions guide selection.

PLS construction and validation

Residual target	Perceived ideology net of WEB, party, incumbency, and year
Direction	Normalized covariance of the embedding with the residual
Scoring	Five folds: estimate the direction outside the scored fold
Inference	Repeat score construction under the null

Random-direction benchmarks and repeated-construction permutation tests answer different questions.

Controls and analysis samples

Specification	Included variables	<i>N</i>
Baseline	Party, WEB, CF score, woman, incumbency, year	1,860
District composition	Baseline + density, education, race, foreign-born share, income, employment	1,855
Party follow-up	Also presidential vote and candidate visibility proxies	Varies

Outcome: BAM perceived ideology. OLS models with HC1 standard errors in Table 2.

The directions overlap only partly

	WEB	Woman	PCA	PLS	Lexicon
WEB	1.00	−0.38	−0.43	0.00	−0.03
Woman	−0.38	1.00	0.28	0.05	0.13
PCA	−0.43	0.28	1.00	0.41	0.46
PLS	0.00	0.05	0.41	1.00	0.47
Lexicon	−0.03	0.13	0.46	0.47	1.00

Pearson correlations across 4,368 candidate-years. Scores oriented toward the people pole.

Only PLS remains in the joint model



All scores entered jointly. Full district controls. $N = 1,855$. 95% HC1 intervals.

Only PLS remains in the joint model

	PCA		PLS		Lexicon		All
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
PCA	-0.103 (0.012)	-0.017 (0.012)					0.008 (0.013)
PLS			-0.194 (0.011)	-0.088 (0.013)			-0.098 (0.014)
Lexicon-only					-0.088 (0.012)	-0.012 (0.011)	0.015 (0.012)
WEB	0.401 (0.029)	0.306 (0.026)	0.412 (0.028)	0.325 (0.026)	0.458 (0.030)	0.314 (0.026)	0.325 (0.027)
CF score	0.054 (0.052)	0.309 (0.045)	0.148 (0.049)	0.324 (0.045)	0.027 (0.053)	0.307 (0.046)	0.325 (0.045)
Republican	2.784 (0.106)	2.298 (0.095)	2.573 (0.099)	2.258 (0.094)	2.752 (0.108)	2.288 (0.095)	2.254 (0.095)
Woman	-0.106 (0.027)	-0.095 (0.025)	-0.105 (0.025)	-0.091 (0.024)	-0.112 (0.027)	-0.096 (0.024)	-0.096 (0.024)
Incumbent	-0.103 (0.024)	-0.047 (0.022)	-0.082 (0.023)	-0.048 (0.022)	-0.112 (0.024)	-0.048 (0.022)	-0.045 (0.022)
District comp.	No	Yes	No	Yes	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	1,860	1,855	1,860	1,855	1,860	1,855	1,855
<i>R</i> ²	0.931	0.945	0.939	0.947	0.931	0.945	0.947

Table: Campaign rhetoric and perceived candidate ideology. OLS estimates with HC1 standard errors in parentheses. The outcome is the BAM estimate of perceived ideology (higher values are more conservative). WEB is the left-right rhetoric score. Each rhetorical score is standardized within party and oriented so that higher values are nearer the people pole; PLS scores are cross-fitted. District composition includes log population density, the percentages with a bachelor's degree, non-Hispanic white, and foreign-born, log median household income, and the shares of employment in production and in agriculture and extraction. For 1,000 random directions in the embedding space, the 95th percentile of the absolute coefficient is 0.046 without district composition and 0.031 with it.

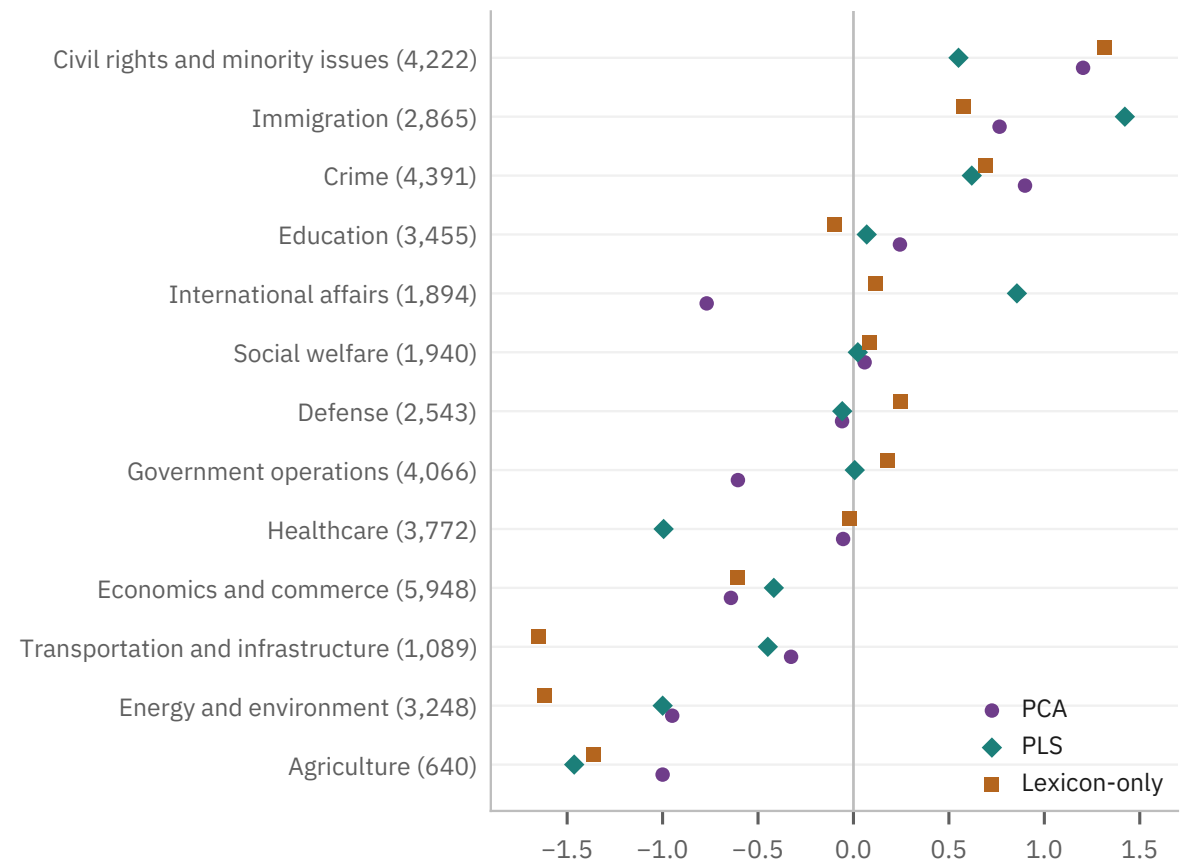
The parties recover different vocabularies

	Democrats	Republicans
Upper pole	muslim, injustices, eviction, detained, brutality	gangs, neighborhood, rents, sheriffs, bail
Lower pole	mechanics, upgraded, installation, engineers	hunting, fishing, salmon, watershed, fracking

Compare the party-specific maps

Cosine between party-specific PLS directions = 0.06

Issue categories agree at the poles



Mean statement score (z): negative = stuff; positive = people

Source: manuscript Figure 6. Counts in parentheses.